



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2019/2020

SUPPLEMENTARY/SPECIAL EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN ANIMAL HEALTH AND PRODUCTION

KAS 106: CELL BIOLOGY

DATE: FRIDAY 15TH JANUARY 2021

TIME: 9.00 A.M. – 12.00 P.M.

INSTRUCTIONS

Answer Question One in Section A and any TWO Questions in Section B

SECTION A: Answer ALL the Questions(5 marks each)

1. State the functions of the following organelles
 - i) Lysosomes
 - ii) Peroxisomes
 - iii) Cilia
 - iv) Flagella
 - v) Centrioles
2. Explain M.J. Schleiden and Theodore Schwans cell theory
3. State the major differences of plant and animal cell
4. Differentiate between the following
 - i) Endocytosis an exocytosis
 - ii) Phagocytosis and pinocytosis
 - iii) Active transport and osmosis
 - iv) Hydrophobic and hydrophilic
 - v) Early prophase and middle prophase
5. Compare and contrast light microscope and electron microscope
6. Briefly describe the fluid mosaic model proposed by Singer and Nicholson in 1972.
7. Compare and contrast chloroplast and mitochondria
8. Describe briefly the functions of plasma membrane of a cell

SECTION B: Answer Two questions (30 marks)

9. Giving example describe the differences between prokaryotic and Eukaryotic cells (15 marks)
10. With a well-illustrated diagram describe the steps involved in cell division. (15 marks)
11. Discuss the phases of mitosis in an animal cell (15 marks)